ARSLAN SHAFIQUE

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RESEARCH PROFILE

Computer Science graduate with hands-on experience in embedded sensing, signal processing and machine-learning applications. Developed ESP32-S3 firmware and device-to-backend communication for an award-winning wearable-monitoring prototype, including real-time audio acquisition and event-triggered processing. Seeking M.Sc. research in dependable multimodal inference and resource-aware edge intelligence.

EDUCATION

B.S. Computer Science (Honors) — University of Central Punjab, Lahore | Oct 2022 – Jul 2026

Official CGPA: 2.63/4.00 | 132 credits completed

Relevant results: Grade A in Final Year Project I & II and Microcontroller Programming; B+ in Deep Learning.

Selected coursework: Deep Learning, Artificial Intelligence, Computer Networks, Data Structures & Algorithms, Probability & Statistics.

RESEARCH & TECHNICAL EXPERIENCE

Smart Baby Band — Cry & Sleep Monitoring System | Final Year Project | Oct 2025 – Jun 2026

Three-member team • Advisor: Mr. Muhammad Ammar Hassan • My responsibility: embedded hardware, ESP32-S3 firmware, sensor integration, and device-to-backend communication.

- Integrated MAX30102, MPU6050, BME280 and INMP441 modules with the ESP32-S3 to acquire physiological, motion, environmental and acoustic data.
- Implemented 16 kHz I^2S audio capture, a 20 ms-frame voice-activity gate (energy, zero-crossing rate and peak criteria) and a three-second circular buffer; separated sensing and networking through FreeRTOS tasks.
- Implemented MQTT telemetry and event-triggered HTTP transfer of captured audio to the backend; connected the device to the team's FastAPI cry-analysis pipeline and Flutter alert workflow.
- Tested the feasibility of on-device TinyML cry detection and selected an edge-gating / backend-classification split to manage limited processing resources.
- Team received the Best Category Project Award at UCP Open House on 6 June 2026 (nearly 300 projects exhibited).

Project code: Smart Baby Band repository

SELECTED MACHINE-LEARNING PROJECTS

Speech Emotion Recognition — CNN-LSTM | RAVDESS | May 2026

- Trained a five-class speech-emotion model using MFCC features, audio augmentation and a CNN-LSTM workflow; analyzed per-class prediction errors.
- Validation results: 86.32% accuracy and 0.85 macro-F1 on the reported validation split. A held-out speaker-independent test has not been established.

SELECTED MACHINE-LEARNING PROJECTS (CONTINUED)

Explainable Speech Emotion Recognition | TESS | 2026

- Explored speech-feature attribution using neural audio classification and visualization; documented limits of inference from the two-speaker TESS dataset.

Project code: Speech-emotion repository

OPEN-SOURCE CONTRIBUTIONS (AI-ASSISTED)

Pylint — PR #11646: Submitted an AI-assisted proposal for an AST-based misplaced-module-dunder checker, with related tests and documentation. View PR

Oppia — PR #27564: Submitted an AI-assisted proposal with targeted question_domain.py tests and a migration-filter correction. View PR

RELEVANT PROFESSIONAL EXPERIENCE

AI Engineering Intern — Techohub Systems (Productix) | Jul – Sep 2026

Contributed Python/FastAPI services, database-connected APIs and AI-assisted application integrations; gained practice in implementing and testing multi-component software workflows.

TECHNICAL & RESEARCH METHODS

Embedded systems: ESP32-S3, FreeRTOS, I^2C/I^2S, real-time audio acquisition, circular buffering, MQTT/HTTP telemetry.

Machine learning: Python, PyTorch, scikit-learn, CNN/LSTM, MFCC/mel-spectral features, classification evaluation.

Reproducible development: Git/GitHub, Jupyter, unit testing, code review, FastAPI, SQL databases.

SELECTED CERTIFICATIONS & TRAINING

• **Embedded TinyML — Edureka.** Relevant to microcontroller-based machine learning and resource-constrained edge inference.

• **IBM / Coursera — Generative AI Engineering with LLMs Specialization (2026):** Seven-course credential track; includes Engineering with Transformers & LLMs, Advanced Fine-Tuning for LLMs, and AI Agents Using RAG and LangChain.

• Udemy — Python for Data Science and Machine Learning Bootcamp (2025).

• Packt — Foundations of LLMs and Deep Learning for Text Analysis (2026).

Credential verification: IBM Credly profile

ENGLISH LANGUAGE

B.S. degree taught in English; university Medium of Instruction (MOI) letter available.